PROJECT: WILL THIS CUSTOMER PURCHASE YOUR PRODUCT?

datalab

Online shopping decisions rely on how consumers engage with online store content. You work for a new startup company that has just launched a new online shopping website. The marketing team asks you, a new data scientist, to review a dataset of online shoppers' purchasing intentions gathered over the last year. Specifically, the team wants you to generate some insights into customer browsing behaviors in November and December, the busiest months for shoppers. You have decided to identify two groups of customers: those with a low purchase rate and returning customers. After identifying these groups, you want to determine the probability that any of these customers will make a purchase in a new marketing campaign to help gauge potential success for next year's sales.

Data description:

You are given an `online_shopping_session_data.csv` that contains several columns about each shopping session. Each shopping session corresponded to a single user.

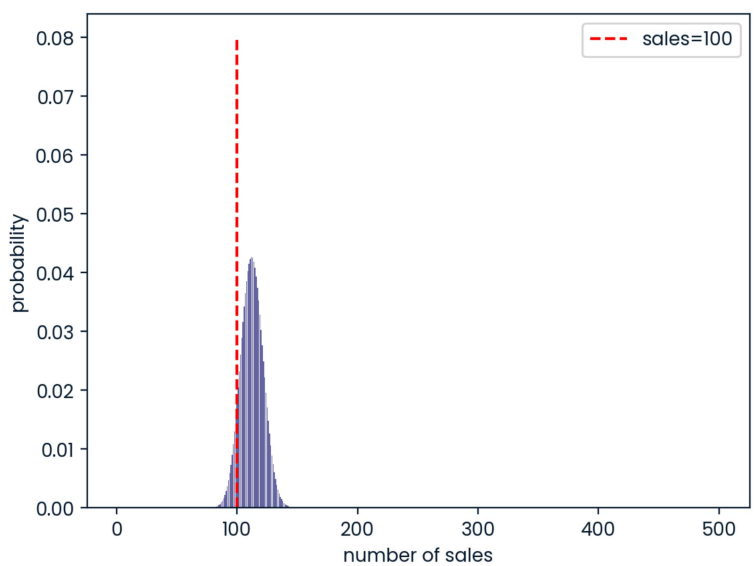
Column	Description
`SessionID`	unique session ID
`Administrative`	number of pages visited related to the customer account
`Administrative_Duration`	total amount of time spent (in seconds) on administrative pages
`Informational`	number of pages visited related to the website and the company
`Informational_Duration`	total amount of time spent (in seconds) on informational pages
`ProductRelated`	number of pages visited related to available products
`ProductRelated_Duration`	total amount of time spent (in seconds) on product-related pages
`BounceRates`	average bounce rate of pages visited by the customer
`ExitRates`	average exit rate of pages visited by the customer
`PageValues`	average page value of pages visited by the customer
`SpecialDay`	closeness of the site visiting time to a specific special day
`Weekend`	indicator whether the session is on a weekend
`Month`	month of the session date
`CustomerType`	customer type
`Purchase`	class label whether the customer make a purchase

```python
# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

# Load and view your data
shopping_data = pd.read_csv("online_shopping_session_data.csv")
shopping_data.head()
```

	S.	Adminis...	Administrative_Duration	Inform...	Informational_Duration	Product...	Pro
0	1	0	0	0	0	1	
1	2	0	0	0	0	2	
2	3	0	0	0	0	1	
3	4	0	0	0	0	2	
4	5	0	0	0	0	10	

Rows: 5 Expand

```python
# Start your code here!
# Use as many cells as you like
# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

# Load and preview data
shopping_data = pd.read_csv("online_shopping_session_data.csv")
shopping_data.head()

## Calculate the online purchase rates during online shopping sessions by customer type for November and December.
# Subset dataframe for November and December data
shopping_Nov_Dec = shopping_data[shopping_data['Month'].isin(['Nov', 'Dec'])]

# Preview to make sure the subset is correct
print(shopping_Nov_Dec.head())

# Make sure we only have November and December data
print(shopping_Nov_Dec['Month'].unique())

# Get session frequency stats by CustomerType and Purchase
count_session = shopping_Nov_Dec.groupby(['CustomerType'])['Purchase'].value_counts()
print(count_session)

# Total number of session by CustomerType
total_new_customer = np.sum(count_session['New_Customer'])
total_returning_customer = np.sum(count_session['Returning_Customer'])

# Total number of purchase by CustomerType
purchase_new_customer = count_session[('New_Customer', 1)]
purchase_returning_customer = count_session[('Returning_Customer', 1)]

# Calculate purchase rates
purchase_rate_new = purchase_new_customer / total_new_customer
purchase_rate_returning = purchase_returning_customer / total_returning_customer

# Therefore, the online purchase for the returning customers is lower than that of the new customers.
purchase_rates = {"Returning_Customer": purchase_rate_returning, "New_Customer": purchase_rate_new}
print(purchase_rates)

## Identify the strongest correlation in total time spent between different types of pages visited by the returning customers
during the months of November and December.
# Calculate correlation with pandas
cor_admin_info = shopping_Nov_Dec['Administrative_Duration'].corr(shopping_Nov_Dec['Informational_Duration'])
cor_admin_product = shopping_Nov_Dec['Administrative_Duration'].corr(shopping_Nov_Dec['ProductRelated_Duration'])
cor_product_info = shopping_Nov_Dec['ProductRelated_Duration'].corr(shopping_Nov_Dec['Informational_Duration'])

print(cor_admin_info)
print(cor_admin_product)
print(cor_product_info)

# Another way to solve this is to use Scipy pearsonr function
cor_admin_info = stats.pearsonr(shopping_Nov_Dec['Administrative_Duration'], shopping_Nov_Dec['Informational_Duration'])

# Another way to solve this is to use Pandas correlation matrix
shopping_Nov_Dec[['Administrative_Duration','Informational_Duration','ProductRelated_Duration' ]].corr()

# Store top correlation
top_correlation = {"pair": ('Administrative_Duration', 'ProductRelated_Duration'), "correlation": cor_admin_product}
print(top_correlation)

## A new campaign for the returning customers will boost the purchase rate by 15%.
## What is the likelihood of achieving at least 100 sales out of 500 online shopping sessions for the returning customers?

# Purchase is a binomial random variable taking the value of either 0 or 1
# We know that the current purchase rate for the returning customers is
print("Current purchase rate for the returning customer:", purchase_rate_returning)

# 15% Increase in this rate would be
increased_purchase_rate_returning = 1.15 * purchase_rate_returning
```

```
print("Increased purchase rate for the returning customer:", increased_purchase_rate_returning)

# First, we find the likelihood of having <100 sales of 500 sessions
# We can find this from binomial cdf
prob_sales_100_less = stats.binom.cdf(k=100, n=500, p=increased_purchase_rate_returning)
print("probability of having <100 sales:", prob_sales_100_less)

# Then, to find the probability of having 100 or more sales is 1-prob_sales_100_less
prob_at_least_100_sales = 1 - prob_sales_100_less
print("probability of having at least 100 sales:", prob_at_least_100_sales)

# Plotting the binomial probability distribution
n_sessions = 500
k_values = np.arange(500) + 1
p_binom_values = [stats.binom.pmf(k, n_sessions, increased_purchase_rate_returning) for k in k_values ]
plt.bar(k_values, p_binom_values)
plt.vlines(100, 0, 0.08, color='r', linestyle='dashed', label="sales=100")
plt.xlabel("number of sales")
plt.ylabel("probability")
plt.legend()
plt.show()
```

```
      SessionID  Administrative  ...        CustomerType  Purchase
5463       5464               1  ...        New_Customer       0.0
5464       5465               3  ...  Returning_Customer       1.0
5467       5468               4  ...  Returning_Customer       0.0
5479       5480               0  ...  Returning_Customer       1.0
5494       5495               0  ...  Returning_Customer       1.0

[5 rows x 15 columns]
['Nov' 'Dec']
CustomerType        Purchase
New_Customer        0.0          529
                    1.0          199
Returning_Customer  0.0         2994
                    1.0          728
Name: Purchase, dtype: int64
{'Returning_Customer': 0.1955937667920473, 'New_Customer': 0.2733516483516483}
0.24468855792839267
0.3898546003206963
0.36712552534442133
{'pair': ('Administrative_Duration', 'ProductRelated_Duration'), 'correlation': 0.3898546003206963}
Current purchase rate for the returning customer: 0.1955937667920473
Increased purchase rate for the returning customer: 0.22493283181085436
probability of having <100 sales: 0.0987778660962733
probability of having at least 100 sales: 0.9012221339037267
```